UNIVERSITY OF CAMBRIDGE

DiS MPHIL

Tuning-Free Plug-and-Play ADMM for Sparse-View CT: A Reinforcement-Learning Approach with Off-the-Shelf Denoisers

Emilia Zabrzanska

Magdalene College

July 2026

Submitted in partial fulfillment of the requirements for the Master of Philosophy in Data Intensive Science

Total page count: 16

Main chapters (excluding front-matter, references and appendix): 8 pages (pp 7–14)

Main chapters word count: 0

Methodology used to generate that word count:

```
$ make wordcount
gs -q -dSAFER -sDEVICE=txtwrite -o - \
   -dFirstPage=6 -dLastPage=11 report-submission.pdf | \
egrep '[A-Za-z]{3}' | wc -w
```

Declaration

I, Emilia Zabrzanska of Magdalene College, being a candidate for Master of Philosophy in Data Intensive Science, hereby declare that this report and the work described in it are my own work, unaided except as may be specified below, and that the report does not contain material that has already been used to any substantial extent for a comparable purpose.

In preparation of this report, I adhered to the Department of Computer Science and Technology AI Policy. [The project required the approved use of {insert name of technology} and such use is acknowledged in the text at the relevant sections.]

Signed:

Date:

Abstract

This project investigates learned priors for low-dose computed tomography reconstruction, with particular focus on stability, regularisation behaviour, and quantitative performance relative to classical reconstruction approaches.

Acknowledgements

Optional acknowledgements go here.

Contents

Chapter 1

Introduction

Chapter 2

Background

2.1 The Sparse-View CT Inverse Problem

$$\boldsymbol{y} = A\boldsymbol{x} + \boldsymbol{\eta}, \tag{2.1}$$

2.2 Classical Reconstruction

2.3 Plug-and-Play Priors

Chapter 3

Related Work

3.1 Learned Reconstruction: A Brief Landscape

3.2 TFPnP: Reinforcement-Learned Parameter Schedules

Table 3.1: Sparse-view CT PSNR (dB) from Wei et al. [2022], Table 5. The paper's headline result against which this project's reproduction is compared in chapter 5.

Method	**5%**	**7.5%**	**10%**
FBP [Herman, 2009]	18.37	16.16	14.32
TV [Sidky and Pan, 2008]	21.47	19.17	16.95
FBPconv [Jin et al., 2017]	23.00	22.94	22.86
RED-CNN [Chen et al., 2017]	23.45	23.11	22.71
LPD [Adler and Öktem, 2018]	23.86	23.44	23.00
RPGD [Gupta et al., 2018]	24.29	23.82	23.38
TFPnP	**24.50**	**24.23**	**23.72**

3.3 Position of This Work

Chapter 4

Design and Implementation

4.1 Pipeline Overview

4.2 Forward Model and ADMM Environment

4.3 DRUNet Denoiser

4.4 Policy Network and Training Algorithm

4.5 Hyperparameters and Divergences from the Paper

Table 4.1: Hyperparameter choices and divergences from Wei et al. [2022].

Parameter	**Paper**	**This work**	**Rationale**
Image size	128×128	512×512	LIDC native res.
Detector pixels	182	724	Scales with image.
Views	30	30	Paper-faithful.
m, N	5, 6	5, 6	Paper-faithful.
η (reward penalty)	0.05	0.05	Paper-faithful.
γ (discount)	0.99	0.99	Paper-faithful.
Batch size	48	8	Memory at 512^2.
n_{grad}	10	4	Memory + π_2 cost.
σ range	(1, 50)	(1, 5)	DRUNet calibration.
μ range	–	(10, 100)	ADMM convergence.
$\text{lr}_{\text{policy}}$	10^{-4}	3×10^{-5}	Stability at 512^2.
$\text{lr}_{\text{critic}}$	5×10^{-5}	10^{-4}	Convention.
LR schedule	halve @ 1600	none	Shorter training.
π_2 warmup	–	5 ep	Stabilises π_2.
π_2 loss scale	–	0.01	Prevents explosion.
π_2 grad-clip	–	0.1	Stability.
π_2 update freq	every	every other	Stability.
Replay buffer	–	5k / 10k	Memory-bound.

4.6 σ/μ Calibration and Domain Mismatch

4.7 Data, Baselines, Metrics

Chapter 5

Evaluation

5.1 Experimental Setup

5.2 Headline Quantitative Results

5.3 Training Dynamics and Policy Behaviour

5.4 Qualitative Reconstructions

5.5 Extensions

Chapter 6

Discussion

6.1 Interpreting the Results

6.2 Depth Beats Breadth

6.3 The σ-Floor Pressure Phenomenon

6.4 DRUNet Over-Smoothing

6.5 Limitations and Caveats

Chapter 7

Conclusion

Bibliography

Jonas Adler and Ozan Öktem. Learned primal-dual reconstruction. *IEEE Transactions on Medical Imaging*, 37(6):1322–1332, 2018.

Hu Chen, Yi Zhang, Mannudeep K. Kalra, Feng Lin, Yang Chen, Peixi Liao, Jiliu Zhou, and Ge Wang. Low-dose CT with a residual encoder-decoder convolutional neural network. *IEEE Transactions on Medical Imaging*, 36(12):2524–2535, 2017.

Harshit Gupta, Kyong Hwan Jin, Ha Q. Nguyen, Michael T. McCann, and Michael Unser. CNN-based projected gradient descent for consistent CT image reconstruction. *IEEE Transactions on Medical Imaging*, 37(6):1440–1453, 2018.

Gabor T. Herman. *Fundamentals of Computerized Tomography: Image Reconstruction from Projections*. Springer Science & Business Media, 2009.

Kyong Hwan Jin, Michael T. McCann, Emmanuel Froustey, and Michael Unser. Deep convolutional neural network for inverse problems in imaging. *IEEE Transactions on Image Processing*, 26(9):4509–4522, 2017.

Emil Y. Sidky and Xiaochuan Pan. Image reconstruction in circular cone-beam computed tomography by constrained, total-variation minimization. *Physics in Medicine and Biology*, 53(17):4777, 2008.

Kaixuan Wei, Angelica Aviles-Rivero, Jingwei Liang, Ying Fu, Hua Huang, and Carola-Bibiane Schönlieb. TFPnP: Tuning-free plug-and-play proximal algorithms with applications to inverse imaging problems. *Journal of Machine Learning Research*, 23(16):1–48, 2022.

Appendix A

Supplementary Material

A.1 Tomosipo Operator Validation

A.2 DRUNet Gaussian Validation